

FRAUD DECISION SUPPORT FOR BANKS AND WALLETS

SafePay MENA

Safer instant payments.
Fewer unnecessary checks.

An AI investigator uses mobile-network evidence to help banks decide when a payment can proceed, needs review, or should stop.

0

external calls on the routine path

36 / 36

synthetic acceptance cases passed

3

markets represented in the demo

Karim Mohamed Abdelnabi
Working prototype / September 2026

Egypt / Saudi Arabia / UAE

PUBLISHED MARKET FACTS

Instant payments are becoming everyday infrastructure.

The opportunity is to protect the moment a customer pays, inside the apps they already use.

— EGYPT

1.5bn

IPN transactions
reported for 2024

Nearly 1.5 billion transactions,
worth approximately EGP 2.9 trillion. [1]

— SAUDI ARABIA

85%

of retail payments
were electronic in 2025

14.6 billion electronic transactions
across national payment systems. [2]

— UNITED ARAB EMIRATES

12.5m+

registered Aani users
reported in April 2026

74 connected financial institutions;
average transfers within 3 seconds. [3]

SafePay's customer is the bank or wallet behind the payment. The consumer keeps their existing app.

VISA STAY SECURE 2025 / SURVEYED ADULTS IN EACH COUNTRY

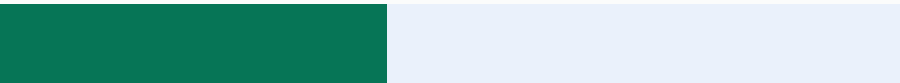
More than 4 in 10 respondents had been scam victims.

Share answering yes when asked whether they had **ever** been the victim of a scam.

EGYPT

43%

of respondents

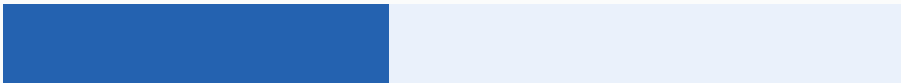


600 adults surveyed

SAUDI ARABIA

43%

of respondents



300 adults surveyed

UNITED ARAB EMIRATES

49%

of respondents



300 adults surveyed

Scams affect customers across all three target markets. Protection must preserve their confidence in everyday payments.

Self-reported lifetime experience. These figures are not annual loss rates, percentages of fraudulent payments, or a measure of SafePay's addressable fraud.

[9] Visa Stay Secure 2025, pp. 5 and 13. Wakefield Research; fieldwork 4-15 Dec 2024. Weighted online survey of 5,800 adults across 17 markets. Bars use a 0-100% scale.

THE LOSS GOES BEYOND THE STOLEN PAYMENT

AED 1 lost to fraud can cost a financial institution AED 4.99.

UAE FINANCIAL SERVICES / SURVEY BENCHMARK

4.99_x

Total cost per dirham lost, including the original loss and associated costs.



LexisNexis / Forrester, July 2023 survey, published 2024.
Institutional survey average; not a SafePay savings multiplier. [4]

- 01

The customer is deceived
A scammer convinces the real account holder to send money. A valid identity can still make an unsafe payment.
- 02

The account is taken over
A changed SIM or device can add evidence that the person making the payment has changed.
- 03

Stolen card details are used
A checkout can look plausible while the session and registered mobile number do not match.

The decision needs both payment context and fresh identity evidence.

ONE DECISION BEFORE THE BANK RELEASES THE PAYMENT

Let normal payments through. Act when the evidence changes.

SafePay screens the context first, investigates elevated risk, and returns a clear action.

FAMILIAR RECIPIENT / ESTABLISHED SESSION

EGP 200

Approve

Routine context stays on the local path. No extra network or AI request.

Customer benefit: no added verification step.

CUSTOMER REPORTS A SUSPICIOUS BANK CALL

SAR 5,000

Hold for review

Even a matching mobile number does not prove the customer is acting freely.

Customer benefit: a chance to stop and reconsider.

UNUSUAL SESSION + SIM AND DEVICE CHANGES

AED 1,200

Block the attempt

Combined warning signals meet the protective policy for this attempt.

Bank benefit: stop a risky release with reasons.

2

Required checks at first setup or a new phone: verify the mobile number and check for a recent SIM change. When required evidence is unavailable, return a retry instead of treating it as clear.

ACTUAL PROTOTYPE / CONNECTED NOKIA SANDBOX

Judges can watch the agent choose evidence in real time.

1 Choose context

2 Run review

3 Inspect outcome

Transaction context

Select a repeatable judge scenario or Nokia sandbox mode.

Market

Egypt

Saudi Arabia

United Arab Emirates

Evidence mode

Connected: Nokia + Gemini

Connected sandbox

Protected Nokia and Gemini access

UNLOCKED

Connected quota is temporarily exhausted. Fixture mode remains available.

Scenario

SIM-swap takeover

An unusual session follows a SIM and device change.

Payment channel

Instant account transfer

Amount

35000

EGP

Recipient relationship

New payee

GEMINI WITH POLICY

Payment blocked

The payment remains stopped for protective fraud review; automated policy never weakens this outcome.

Connected investigation

COMPLETE

5 NOKIA | Evidence received

Nokia sandbox returned SUCCESS

Number Verification | SUCCESS | HTTP 200 | 1737.03 ms

6 GEMINI | API selected

Gemini selected Device Swap

Device Swap

7 NOKIA | Evidence received

Nokia sandbox returned SUCCESS

Device Swap | SUCCESS | HTTP 200 | 224.47 ms

8 POLICY | Policy decision

Deterministic policy validated the agent proposal and observed evidence

Nokia responses are from documented simulator subjects. Bank context remains synthetic and no payment moves.

Pre-call score

80

Final risk score

90

Agent turns + CAMARA tools

7

Why this happened

1

See the payment action first

Approve, hold, block, or retry stays prominent. The user does not need to interpret technical output.

2

Follow orchestration as it happens

The timeline separates local screening, Gemini selection, Nokia evidence and enforced policy.

3

Inspect only when needed

Judges and analysts can expand returned observations and the bounded agent trace.

Live walkthrough: zero-call routine payment → connected takeover review.

Actual 10 September app capture: protected local connected run.
Gemini selected three checks; Nokia returned three sandbox successes.

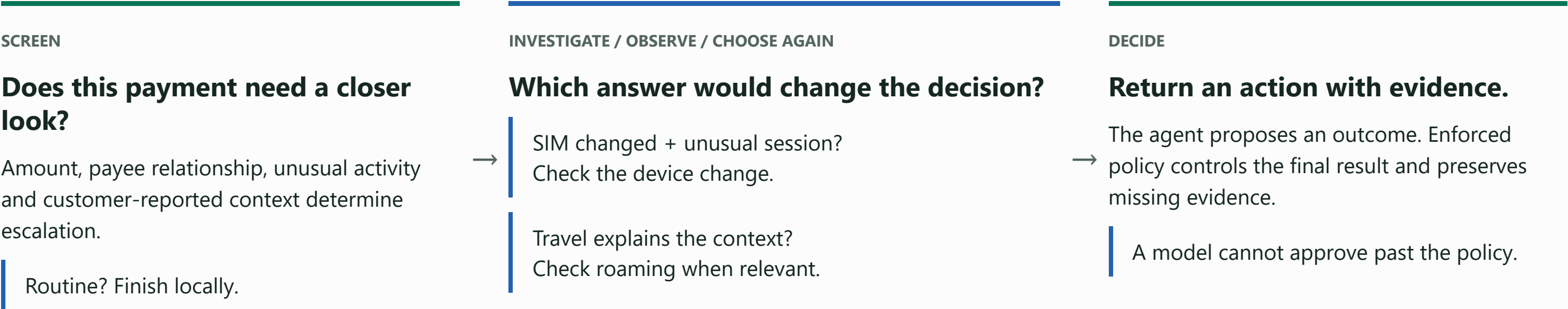
Connected capture used synthetic bank facts and Nokia simulator subjects; no production payment or subscriber was accessed. Desktop and mobile journeys were browser-tested. [6, 7]

06 / 19

ASK THE NEXT USEFUL QUESTION

The agent chooses which evidence to request, then adapts.

Open Gateway lets an approved bank workflow ask the mobile network for relevant risk signals.



0 calls

Routine payments bypass AI and network tools.

Up to 5 tools

Bounded investigation; stop when evidence is sufficient.

2 mandatory checks

First setup and new-device trust follow an enforced rule.

MEASURED SOFTWARE BEHAVIOR + RECORDED NOKIA SANDBOX EVIDENCE

A working prototype with repeatable acceptance tests.

36 / 36

synthetic acceptance cases passed

12 journeys across Egypt, Saudi Arabia and the UAE.

114 / 114

automated tests passed in the 10 September QA run

88 backend / adapter tests and 26 browser tests.

5 APIs

integrated, with recorded Nokia sandbox successes

SIM, number, device, roaming and reachability.

We test the difficult paths too

Provider evidence unavailable

Retry

Clean identity + suspicious transfer

Hold

Expected travel with clear evidence

Approve

Real integration, explicit scope

A verified live takeover run used **3 context-eligible Nokia tools**, across **4 Gemini turns**, and returned **BLOCK**.

These results establish control flow and integration. Fraud-loss reduction and production performance require a bank pilot.

[6] Automated QA, 10 Sep 2026; 36-case acceptance rerun during deck build. [7] Nokia verification, 8-10 Sep 2026. Protected connected mode also preserves explicit fixture fallback.

EXTERNAL CATEGORY EVIDENCE / SAFEPLAY DIFFERENTIATION

Network-informed intervention has a business precedent.

EXTERNAL UK PILOT / FICO + JERSEY TELECOM

44%

lower scam losses

55%

fewer false positives

Vendor-reported results from the first bank pilot, published September 2024. [5]

That deployment used Scam Signal and customer intervention. SafePay has different signals and has not measured these outcomes.

SafePay's implementation focus

Selective investigations

Routine payments avoid extra calls; elevated risk prompts the agent to select relevant evidence.

Multiple journeys, one decision workflow

New-phone setup, takeover, card misuse, suspicious transfers and travel use a shared evidence model.

Visible reasons and controlled decisions

Policy limits agent actions, records missing evidence, and supports repeatable evaluation.

The commercial position: add network evidence to a bank's existing fraud workflow and measure the incremental value.

ILLUSTRATIVE MONTHLY MODEL / USD / ASSUMPTIONS TO VALIDATE

At 25% lower losses, a bank retains \$12,000 a month.

Example baseline: **\$80,000/month** in fraud losses in the payment portfolio covered by the pilot.

\$20k

avoided losses

\$80,000 baseline × 25%
assumed reduction

\$8k

SafePay monthly fee

Proposed platform + usage
at the volume on the next slide

\$12k

monthly benefit
after SafePay fees

Before bank implementation
and incremental operating costs

10%

loss reduction covers the fee.
\$8,000 ÷ \$80,000 baseline losses.

\$144,000 annualized benefit at the same monthly assumptions.
No SMS savings, recovery-cost multiplier or conversion uplift included.

[8] Proposed commercial model, not customer results. The 25% reduction is an unvalidated scenario, not derived from the FICO result. Deployment costs raise the actual break-even point. Sensitivity: slide 14.

PROPOSED PRICING / ILLUSTRATIVE DIRECT COSTS / USD PER MONTH

Recurring platform revenue, plus usage when evidence is needed.

\$3,000

monthly platform fee

+

\$0.05 × 100,000

billable investigations per month

=

\$8,000

monthly customer revenue

Cost of serving one modeled customer

Network: 100k × 2.5 checks × \$0.01	\$2,500
AI budget: 100k × \$0.005 per investigation	\$500
Hosting + direct support allocation	\$1,000
Total modeled direct cost	\$4,000

Network prices and AI budgets are assumptions, not vendor quotes.
Paid retries and auth charges must fit the blended cost or be added.

50%

modeled gross margin



\$4k direct costs / \$4k gross profit

Assumes 1 million total decision events/month: 900k routine and 100k evidence investigations, including setup and new-device events. Mix is not yet measured.

[8] One billable investigation is one evidence workflow, not each CAMARA call. Includes mandatory enrollment workflows. Gross profit excludes R&D, sales, general overhead and tax. Scope and overage terms require pilot negotiation.

SELL TO ONE INSTITUTION, REACH ITS EXISTING CUSTOMERS

Start with one bank or wallet. Repeat the integration.

01 Win a focused bank or wallet pilot

Buyer: head of fraud or digital banking. Begin with new-device trust and suspicious payment review.

02 Work with an operator or aggregator

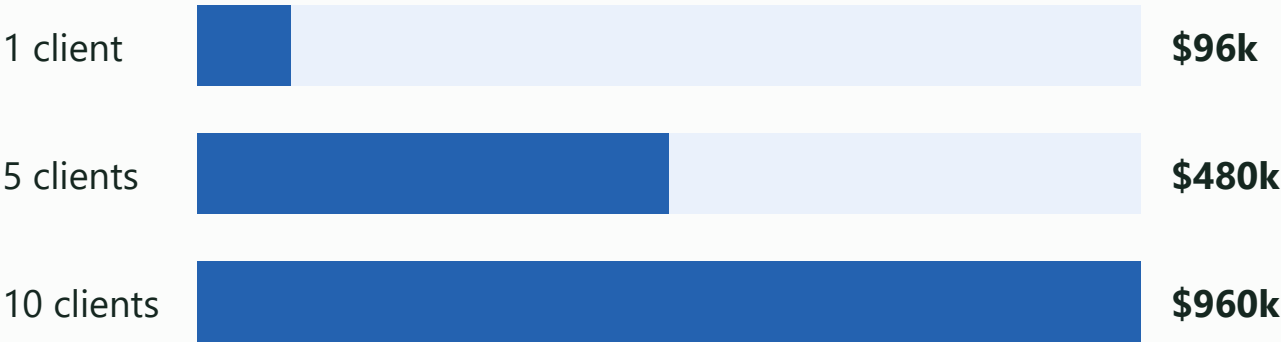
Confirm API coverage, permission flows and wholesale terms for the first target market.

03 Expand after measured value

Use the same decision interface with country-specific bank policy. Egypt, Saudi Arabia and UAE are demo contexts today.

ILLUSTRATIVE SCALE / NOT SIGNED REVENUE

Annual recurring revenue at \$8k/month



Each client is assumed to have the volume and pricing on slide 11. These are scenarios, not a sales forecast.

Operator opportunity: 250,000 modeled network checks per client per month, creating \$2,500 in provider spend at the assumed blended price.

[8] Scale arithmetic: clients × \$8,000 × 12. Provider spend can include an aggregator share; it is not operator net revenue. Regional production coverage and commercial partnerships remain to be secured.

THE ASK

Help us prove the economics in a controlled bank pilot.

1 bank

or wallet partner

A defined payment portfolio,
historical loss labels and a baseline.

1 operator

or network API partner

Confirmed coverage, consent flow,
pricing and service expectations.

90 days

proposed pilot window

Integrate, evaluate in shadow mode,
then test a controlled intervention.

Value target

Test the 25% loss-reduction scenario against the
bank's baseline.

Experience guardrail

No material increase in legitimate-payment
abandonment.

Commercial gate

Validate provider availability, latency, investigation
cost and margin.

Keep everyday payments easy.

Give risky payments the evidence they need.

MAKE THE ASSUMPTIONS TESTABLE

The pilot must prove both savings and cost discipline.

Bank benefit depends on reduced losses

\$80k covered monthly losses; \$8k monthly SafePay fee.

Loss reduction	Avoided losses	After fee
0%	\$0	-\$8,000
10%	\$8,000	\$0
25% / base case	\$20,000	\$12,000
40%	\$32,000	\$24,000

Implementation, added reviews and false-positive losses reduce the actual benefit. The AED 4.99 multiplier is not applied.

SafePay margin depends on network cost

250k checks; \$500 AI; \$1k hosting/support; \$8k revenue.

Blended cost / check	Direct cost	Gross margin
\$0.005	\$2,750	65.6%
\$0.010 / base case	\$4,000	50.0%
\$0.020	\$6,500	18.8%
\$0.030	\$9,000	-12.5%

At \$0.01/check, 4 checks per investigation instead of 2.5 reduce margin to 31.25%. AI cost includes all turns per investigation.

Decision rule for a pilot: net bank benefit = avoided losses – SafePay fee – added bank costs.
Break-even with bank costs: (\$8,000 + monthly added costs) ÷ \$80,000. Prices, volumes and reduction rates are assumptions.

GLOBAL MERCHANT BENCHMARK / NOT A MENA FRAUD-LOSS BREAKDOWN

Several fraud types can affect the same business.

Share of surveyed merchants reporting each attack in the previous 12 months. [10]

Reported attack type	Merchants reporting it	Relation to SafePay's use cases
Real-time payment fraud	45%	Relevant to payment review; the category includes more than the scams modeled in our demo.
Phishing / pharming / whaling	42%	Credential deception can lead to unsafe transfers or takeover. SafePay does not detect phishing content.
Card testing	33%	Testing stolen card details is related to card misuse. This is not the share of all card leakage or CNP fraud.
Account takeover	26%	SIM and device changes can contribute evidence. SIM swap is one possible takeover mechanism.

SIM-swap fraud: a documented attack mechanism, but no reliable MENA share was verified in this review. We do not assign it an invented percentage. [11]

Categories overlap and do not add to 100%. These percentages measure merchants affected, not fraud incidents, losses or SafePay coverage.

SOURCE TRAIL / REVIEWED 10 SEPTEMBER 2026

Every number has a source, a measurement or an assumption.

[1] Egypt payment scale

[Central Bank of Egypt / IPN and InstaPay announcement](#)

30 Dec 2024. Nearly 1.5bn IPN transactions; EGP 2.9tn in 2024. CBE's indexed site excerpt confirms these figures; the full page rejected automated access.

[3] UAE instant-payment adoption

[Al Etihad Payments / Aani operational update](#)

10 Apr 2026. More than 12.5m registered users, 74 financial institutions; average transfers within 3 seconds.

[5] External evidence for network-informed intervention

[FICO and Jersey Telecom / original pilot announcement](#)

18 Sep 2024. First UK pilot bank reported 44% lower scam losses and 55% fewer false positives. Vendor report; different product and signals.

[8] Commercial model

Internal illustrative model, 9 Sep 2026. Inputs and sensitivity are on slides 10-12 and 14. No signed pricing, measured bank ROI or acquired-customer count is asserted.

[2] Saudi electronic payments

[SAMA / E-Payments Account for 85% of Total Retail Payments in 2025](#)

12 Apr 2026. Retail e-payment share: 85%; electronic transaction volume: 14.6bn. This is not sarie-only volume.

[4] Cost of fraud

[LexisNexis Risk Solutions / True Cost of Fraud EMEA](#)

Published 2024; Forrester survey July 2023. UAE financial-services total cost multiplier: 4.99. EMEA sample: 541 decision-makers; UAE subset size not stated.

[6, 7] SafePay prototype evidence

[QA report / Nokia sandbox verification](#)

8-10 Sep 2026. 114 automated tests; 36 synthetic acceptance cases; recorded successes for 5 integrated APIs. Rate-limit handling is documented.

Metric definitions

An investigation is one workflow; a check is one logical network capability. Gross margin = (revenue - direct costs) / revenue. Risk scores are policy scores, not fraud probabilities.

Additional fraud sources: [9] Visa Stay Secure 2025, pp. 5/13 (slide 3); [10] Global eCommerce Payments & Fraud Report 2025, p. 23 (slide 15); [11] GSMA SIM-swap advice (slide 15). Direct source links appear on those slides.

SOLO PROJECT

Karim Mohamed Abdelnabi

Project creator / Product and prototype development

Building SafePay MENA

Karim is the sole team member behind SafePay MENA. He developed the concept around payment fraud in Egypt, Saudi Arabia and the UAE, and brought it into a working web prototype with agent-selected telecom evidence.

His work on this project connects the customer experience, fraud-review workflow, technical implementation and business proposal.

Product and customer experience

Fraud use cases, payment-review journeys and a clear experience for users and judges.

Prototype and integration

Gemini orchestration, Nokia CAMARA integration, policy controls, testing and the Render deployment.

Business and presentation

Regional problem research, proposed pricing, pilot economics and the submission narrative.

Next-stage partners: a bank or wallet for payment integration and a network API provider for production access.

ORGANIZER GUIDE MAPPED TO THE WORKING PROTOTYPE

Three guide-listed tools power the connected demo.

GUIDE SECTION 3 / PAGE 3

Google AI Studio (Gemini)

Gemini API function calling selects relevant evidence tools, consumes their results and proposes the next action.

Used directly from the Python backend.

GUIDE SECTION 5 / PAGE 5

Nokia Network as Code

CAMARA sandbox access supplies SIM Swap, Number Verification, Device Swap, Roaming and Reachability evidence.

Accessed through the RapidAPI gateway.

GUIDE SECTION 6 / PAGE 5

Render

Hosts the FastAPI service and web interface, with provider credentials configured on the server.

Public demo with protected connected access.

Orchestration approach: local screening → Gemini tool request(s) → validated Nokia checks → observations returned to Gemini → enforced policy decision.

Custom bounded Python loop: tool eligibility, request limits and failure handling stay outside the model. No LangGraph, CrewAI, ADK or vector database is used.

Working demo: safepay-mena-demo.onrender.com

TECHNICAL ARCHITECTURE AND EVIDENCE BOUNDARIES

Adaptive investigation. Enforced policy. Inspectable outcomes.

01 / INTERFACE

Web app + FastAPI

Session, setup, payment review, cancellation and evaluation.

Bank facts and payment release are simulated.

02 / RISK + AGENT

Local screen + Gemini

Routine bypass. Gemini can request one or several tools per turn; the backend validates and executes each check.

No scenario labels sent to the model.

03 / NETWORK ACCESS

Nokia Network as Code

CAMARA adapter; fixed hosts, validated responses, bound OAuth state.

Approved simulator subjects only.

04 / FINAL DECISION

Deterministic policy

Evidence-based disposition, missing-data handling and stored result.

In-memory SQLite demonstration store.

Five integrated network capabilities

SIM Swap / Number Verification / Device Swap

Roaming / Reachability

Number Verification checks the number association. It does not prove payment intent, card ownership or freedom from coercion.

Before a production pilot

Provider permission and consent, production identity controls and quotas, durable audit storage, bank integration, calibrated thresholds and measured service levels.

Hosted takeover recording, 11 September: 4 model requests and 3 successful Nokia checks; final decision BLOCK. This is one sandbox observation, not a performance benchmark.